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Spatial-temporal patterns and influencing factors for hemorrhagic fever with renal syndrome: A 16-year national surveillance analysis in China

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ABSTRACT

Background: China is confronted with the significant menace posed by hemorrhagic fever with renal syndrome (HFRS). Nevertheless, the long-term spatial-temporal variations, regional prevalence patterns, and fundamental determinants' mechanisms for HFRS remain inadequately elucidated.
Methods: Newly diagnosed cases of HFRS from January 2004 to December 2019 were acquired from the China Public Health Science Data repository. We used Age-period-cohort and Bayesian Spacetime Hierarchy models to identify high-risk populations and regions in mainland China. Additionally, the Geographical Detector model was employed to quantify the determinant powers of significant driver factors to the disease.
Results: A total of 199,799 cases of HFRS were reported in mainland China during 2004–2019. The incidence of HFRS declined from 1.93 per 100,000 in 2004 to 0.69 per 100,000 in 2019. The incidence demonstrated an inverted U-shaped trend with advancing age, peaking in the 50–54 age group, with higher incidences observed among individuals aged 20–74 years. Hyperendemic areas were mainly concentrated in the northeastern regions of China, while some western provinces exhibited a potential upward trend. Geographical detector model identified that the spatial variations of HFRS were significantly associated with the relative humidity ($Q = 0.36$), forest cover ($Q = 0.26$), rainfall ($Q = 0.18$), temperature ($Q = 0.16$), and the surface water resources ($Q = 0.14$).
Conclusions: This study offered comprehensive examinations of epidemic patterns, identified high-risk areas quantitatively, and analyzed factors influencing HFRS transmission in China. The findings may contribute to the necessary implementations for the effective prevention and control of HFRS.

1. Introduction

Hemorrhagic fever with renal syndrome (HFRS) is a zoonotic disease caused by varying serotypes of Hantavirus (HV), such as Hantaan virus (HTNV), Seoul virus (SEOV), Puumala virus (PUUV), and Dobrava virus (DOBV) [1]. Humans were infected with HFRS primarily through contact with excrement and secretions from infected animals [2]. The global annual incidence of HFRS is estimated to be approximately 60,000–150,000 cases [3], spreading across numerous countries globally. China accounts for 70–90% of the reported cases worldwide [4], with an annual incidence ranging from 20,000 to 50,000 cases, positioning China as the nation bearing the most substantial burden of HFRS globally [5]. China has classified HFRS as a Class B infectious disease for management since 1956 in efforts to strengthen epidemic control. In 2004, a nationwide direct reporting system was established for surveillance, facilitating real-time monitoring of HFRS outbreaks across the country through individual case reports. Furthermore, China has initiated a vaccination program targeting the 16 to 60-year-old population in key regions since 1994. In 2008, a nationwide immunization campaign for HFRS was implemented in prioritized areas to enhance population immunity coverage. Presently, China has successfully implemented preventive and control measures for HFRS, leading to a declining trend in national case numbers [6]. However, cases have been reported in nearly all provinces across the country, showcasing significant regional

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